

ADRAR, Algeria

WMO: 606200

Lat: 27.838N

Lon: 0.186W

Elev: 919

StdP: 14.21

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.7	40.6	5.1	7.4	56.2	8.7	8.9	58.6	34.2	59.8	31.0	60.1	10.4	30	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.5	116.8	72.1	115.1	71.4	113.5	70.8	74.6	104.8	73.0	104.4	71.5	104.4	12.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	98.2	88.1	62.4	87.2	88.6	59.1	77.5	90.4	38.6	105.0	37.1	104.4	35.7	104.1	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 31.7	28.6	25.5	DB	32.7	119.2	2.9	1.1	30.6	120.0	29.0	120.7	27.3	121.3	25.3	122.1	
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	78.3	56.1	61.1	69.1	77.4	86.3	94.4	100.0	98.3	91.8	80.3	66.4	57.8
	DBStd	16.25	5.02	6.15	6.63	6.10	6.16	4.27	3.34	3.37	5.24	5.95	6.20	4.98
	HDD50	13	7	2	0	0	0	0	0	0	0	0	0	4
	HDD65	732	280	136	33	1	0	0	0	0	0	0	52	230
	CDD50	10357	195	313	593	822	1124	1331	1550	1497	1254	940	493	245
	CDD65	5601	3	27	161	373	659	881	1085	1032	804	475	95	6
	CDH74	94192	91	490	2339	5273	10474	15632	20318	18836	13249	6163	1199	128
	CDH80	66070	6	113	1032	2978	6981	11498	15868	14393	9257	3514	415	15
(14) Wind	WSAvg	13.4	12.5	13.6	14.2	14.5	14.4	13.4	14.2	13.6	12.4	12.5	12.1	12.6
(15) Precipitation	PrecAvg	0.8	0.1	0.1	0.0	0.2	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0
	PrecMax	3.3	0.8	0.7	0.3	3.3	0.3	0.2	0.1	0.6	0.7	0.9	0.7	0.4
	PrecMin	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.8	0.2	0.2	0.1	0.7	0.1	0.0	0.0	0.1	0.2	0.2	0.2	0.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.5	88.1	98.5	105.8	111.4	116.6	118.8	117.0	113.2	106.2	94.7	82.0
		MCWB	54.5	57.3	62.6	62.7	68.6	70.8	72.2	70.5	70.6	68.6	61.6	57.6
	2%	DB	76.7	82.8	93.3	100.7	109.1	114.5	117.9	115.2	111.4	101.8	89.2	77.2
		MCWB	53.2	55.5	60.0	64.4	67.8	70.5	72.2	71.1	70.4	66.8	61.1	55.7
	5%	DB	73.4	79.6	89.6	97.2	106.1	112.6	116.5	114.4	109.5	98.6	84.5	74.3
		MCWB	51.7	54.3	58.9	63.1	66.7	70.1	72.2	71.0	69.9	66.0	59.4	54.5
	10%	DB	70.2	76.6	85.9	94.1	103.6	109.9	114.9	112.8	107.4	95.3	80.8	71.4
		MCWB	50.4	53.0	57.5	62.0	66.5	69.1	71.5	70.7	69.6	64.8	58.3	52.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.0	60.4	64.5	70.0	71.9	73.1	75.4	78.2	77.0	73.0	66.0	61.6
		MCDB	74.4	77.5	79.1	88.1	100.3	106.4	112.0	107.0	101.2	96.1	83.5	75.9
	2%	WB	54.9	57.2	60.9	66.2	68.8	71.4	73.7	74.5	74.1	69.9	63.1	57.7
		MCDB	72.2	77.1	83.0	88.5	98.3	106.7	111.0	103.3	100.6	93.4	83.8	73.2
	5%	WB	52.9	55.1	58.8	63.2	66.7	69.6	71.9	72.7	72.3	67.9	60.8	55.5
		MCDB	70.9	75.5	81.8	88.4	96.6	104.6	110.7	103.9	99.5	91.9	80.7	71.8
	10%	WB	50.8	53.4	56.8	60.8	64.9	68.0	70.2	70.9	70.5	66.2	58.6	53.7
		MCDB	68.9	73.6	80.5	86.2	95.7	103.8	109.7	105.4	98.3	90.9	78.3	69.6
(35) Mean Daily Temperature Range	5% DB	MDBR	27.8	28.1	28.8	28.9	29.0	29.6	29.5	27.9	27.0	26.7	26.8	26.8
		MCDBR	30.7	30.7	32.5	32.1	31.1	31.3	30.0	29.3	29.5	29.3	29.6	29.1
		MCWBR	15.9	14.4	13.7	12.9	12.6	12.4	11.5	10.5	12.3	13.0	14.4	15.3
	5% WB	MCDBR	28.5	27.5	26.2	25.5	27.1	28.2	26.3	25.3	25.4	25.8	27.0	27.5
		MCWBR	16.2	14.5	13.7	13.4	12.9	13.0	11.4	11.3	12.3	12.8	15.4	15.8
(40) Clear-Sky Solar Irradiance	taub		0.376	0.419	0.512	0.549	0.609	0.641	0.675	0.658	0.626	0.548	0.426	0.385
	taud		2.071	1.945	1.723	1.662	1.571	1.523	1.477	1.527	1.588	1.720	1.964	2.066
	Ebn at Noon		268	266	248	244	229	221	213	215	216	223	248	259
	Edh at Noon		44	55	73	80	88	91	95	90	82	68	49	43
(44) All-Sky Solar Radiation	RadAvg		1353	1623	1943	2175	2309	2434	2414	2241	1938	1643	1398	1235
	RadStd		62	76	87	70	102	75	37	63	81	87	55	50

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page